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MILITARY SPECIFICATION

VALVE, DIRECTION, FIRE EXTINGUISHING SYSTEM,
TYPE MC-1

1. SCOPE

1.1 This specification covers one type of fire extinguishing system direction valve, designated Type MC-1.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-576	Box, Wood, Cleated, Veneer Paper
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Set-Up, Paperboard

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-B-4394	Bromochloromethane, Technical
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-C-6386	Container, Aircraft Fire Extinguisher, Bromochloromethane, Type B-1A
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-L-10547	Liners, Case, And Sheet, Overwrap: Water-Vaporproof Or Waterproof Flexible

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property

FSC 1680

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MIL-STD-143	Specifications And Standards, Order Of Precedence For The Selection Of
MS33586	Metals, Definition Of Dissimilar
MS33649	Bosses, Fluid Connection-Internal Standard Thread

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The valve furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

* 3.2 Selection of specifications and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph:

3.2.1 Standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings or in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.3 Materials.

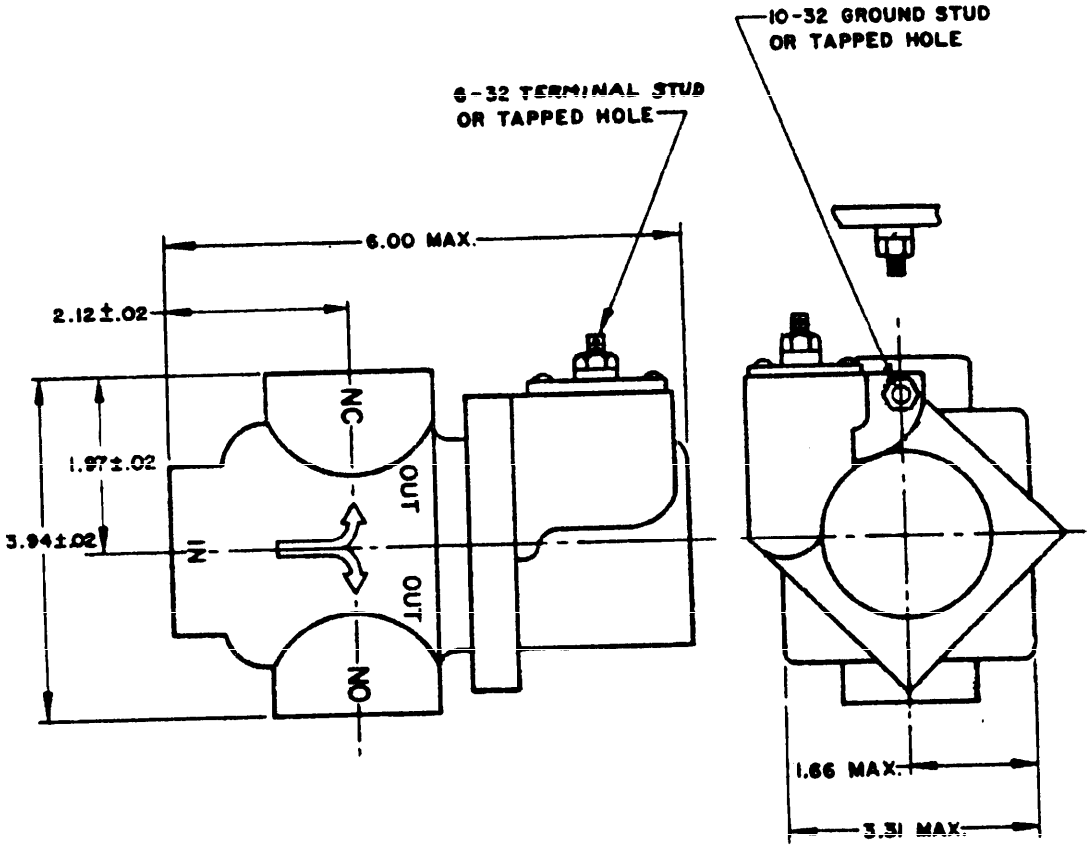
3.3.1 Metals. Metals used in the construction of the valve shall be corrosion resistant or suitably protected to resist corrosion during normal service life.

* 3.3.2 Dissimilar metals. The use of dissimilar metals, especially brass, copper, or steel in intimate metal-to-metal contact with aluminum or aluminum alloys, shall be avoided whenever practicable. Dissimilar metals are defined in MS33566. Magnesium shall not be used in the construction of the valve.

3.3.3 Protective treatment. When materials are used in the construction of the valve that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The valve shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibration, and other conditions incident to shipping, storage, installation, and service. The valve shall conform to Figure 1.

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NOTE:
INLET AND OUTLET PORTS SHALL CONFORM TO MS33649

DIMENSIONS IN INCHES

FIGURE 1. VALVE

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*. 3.4.1 Inlets and outlets. The valve shall have one inlet and two outlets each of which shall be threaded in accordance with MS33649 for 1-1/4 inch outside diameter (OD) tubing. The configuration and dimensions of the inlet and outlets shall be as shown on Figure 1.

* 3.4.2 Actuation. The valve shall be electrically actuated for discharge in one direction and no electrical actuation shall be required for discharge in the alternate direction. The valve shall be designed so that the complete agent discharge is directed in the desired manner without the necessity of the electrical switch being held in the on position throughout the discharge, The agent or pressurizing gases shall hold the valve in the selected direction until the discharge is completed. The actuating voltage shall be from 18 to 29 volts direct current (DC). The power terminal shall be a 6-32 National Coarse NC-2A stud or tapped hole and the ground terminal shall be a 10-32 National Fine NF-2A stud or tapped hole.

* 3.4.3 Restriction to flow. The valve shall be designed to offer the least possible restriction to flow of the extinguishing agent. The addition of the valve to a given fire extinguishing system manifold shall not increase tne discharge time of the system by more than 10 percent. To determine conformance to this requirement, the valve shall meet the test specified in 4.3.3.3.

3.4.4 Ventilation of system manifold. The valve shall have a normally open position so that the complete system is open to the atmosphere, or shall be constructed so that the direction valve vents the discharge tubing upstream of the valve to the atmosphere except during the discharge of the agent.

3.5 Performance. The valve shall be capable of satisfactory operation under the following conditions:

a. Temperatures ranging from -54° to +71° Centigrade (C).

b. Relative humidity up to 100 percent including conditions wherein condensation takes place in the form of both water and frost.

c. Exposure to salt sea atmosphere.

d. Vibration incident to service use.

e. Fungus growth as encountered in tropical climates.

f. Sand and dust particles as will be encountered in desert areas.

g. Corrosion resistance to bromochloromethane.

3.5.1 Flow direction. The valve shall be capable of directing the complete flow of agent in the desired direction when placed not more than 1 foot from the agent container. The container and valve shall be actuated simultaneously. No time delay or other arrangement shall be used that would require the actuation of the valve before discharging the agent from the container. The container shall conform to MIL-C-6386. When the agent is completely discharged, the valve shall return to its original position.

3.5.2 Mounting. The valve shall be capable of proper operation when mounted in any position in the discharge manifold.

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3.5.3 Structural strength.

* 3.5.3.1 Proof pressure. The valve shall be capable of withstanding a proof pressure of 400 pounds per square inch without evidence of leakage, failure, distortion, or permanent set.

3.5.3.2 Burst pressure. The valve shall be capable of withstanding a minimum burst pressure of 725 pounds per square inch.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of MIL-D-1000.

* 3.7 Weight. The weight shall be kept as low as possible, consistent with good design, and shall not exceed 2.6 pounds.

3.8 Finishes and protective coatings.

3.8.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625.

3.8.2 Steel parts. Wherever practicable, steel parts that are subject to corrosion shall be cadmium plated in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.9 Operation marking.

3.9.1 Direction of flow. The valve shall be clearly and permanently marked on the external surface to indicate the direction of flow through the valve. Decalcomanias shall not be considered permanent marking.

3.9.2 Terminals. The terminals shall be clearly and permanently marked on the external surface to differentiate between the power and ground connections.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

* 3.11 Workmanship.

* 3.11.1 General. The valve, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

* 3.12 Screw threads. Screw threads shall be in accordance with MIL-S-7742.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of valves shall be classified as follows:

a. Qualification tests: Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

b. Quality conformance tests. Quality conformance tests are those tests accomplished on valves manufactured and submitted for acceptance under contract.

4.3 Qualification tests.

4.3.1 Prior qualification. The valve which this specification covers shall, prior to acceptance, pass the qualification tests specified herein. If the valve is later modified in any way, the modified form shall also be subjected to and pass the test.

4.3.2 Qualification test sample. Qualification test samples shall consist of two valves. Samples shall be appropriately identified with the manufacturer's own part number and any additional identification required by the authorizing letter.

4.3.3 The qualification test shall consist of the following tests and those specified under the acceptance and sampling tests.

4.3.3.1 Environmental tests. Each valve shall be subjected to the following environmental tests in accordance with the specified procedures of MIL-E-5272.

4.3.3.1.1 High temperature. The high temperature test shall be in accordance with procedure I.

4.3.3.1.2 Low temperature. The low temperature test shall be in accordance with procedure I for a period of 72 hours.

4.3.3.1.3 Humidity. The humidity test shall be in accordance with procedure I.

4.3.3.1.4 Salt spray. The salt spray test shall be in accordance with procedure I for a period of 100 hours.

4.3.3.1.5 Vibration. The vibration test shall be in accordance with procedure I.

4.3.3.1.6 Fungus. The fungus test shall be in accordance with procedure I.

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4.3.3.1.7 Sand and dust. The sand and dust test shall be in accordance with procedure I.

4.3.3.2 Corrosion resistance to bromochloromethane. The valve shall pass a corrosion test which consists of 15 discharges using CB bromochloromethane saturated with 2 percent of water by weight. The bromochloromethane shall conform to MIL-B-4394. The discharges shall extend over a minimum period of 5 weeks with no two discharges run within 44 hours of each other. Consecutive discharges shall be run in alternate directions through the valve. There shall be no purging of the valve during this test. The valve shall operate satisfactorily for each discharge.

4.3.3.3 Restriction to flow test. The discharge time, through a simulated fire extinguishing system, shall be recorded to the nearest one-tenth of a second. The system shall consist of a Type B-1A-6 container conforming to MIL-C-6386, 5 feet of 1-1/4 inch OD tubing, a valve, and a suitable discharge nozzle. The extinguishing system then shall be modified by replacing the valve with a standard 90°, 1-1/4 inch connector. The discharge time of the system, with the direction valve installed, shall not be greater than 110 percent of that of the system with a 90° connector installed. The Type B-1A-6 container may be charged with water for the purpose of this test. The method used to measure the discharge time shall be approved by the procuring activity. The discharge time shall be approved by the procuring activity. The discharge time of each discharge shall be determined in an identical manner.

4.3.3.4 Burst pressure test. The valves selected for sampling test shall be hydrostatically tested to 725 pounds per square inch gage to determine proper strength of parts and materials when subjected to burst pressure. The pressure shall be applied through the inlet with the outlets capped. Pressure shall be applied at least two times for a period of 1 minute. Plastic deformation and malfunction as a result of this test shall not be cause for rejection of the valve. Fracture of the valve housing during this test shall be cause for rejection of the valve.

* 4.4 Quality conformance tests.

4.4.1 Individual tests. Each valve shall be subjected to the following tests.

4.4.1.1 Examination of product. The valve shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.4.1.2 Actuating voltages of 18 and 29 volts DC shall be applied to each valve. The valve shall function correctly for each voltage.

4.4.2 Sampling tests. Valves shall be selected at random from each lot in the quantities indicated below and subjected to the following tests.

Number of Valves in Lot	Number of Sample Valves to be Selected
1 - 25	1
26 - 100	2
101 - 500	3
501 - or more	4

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4.4.2.1 Lot. A lot shall consist of valves manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.4.2.2 When sampling tests are specified on a number of items that are selected as representative of a certain lot, and one or more of this number fail to meet the specified tests, acceptance of all items shall be withheld until the extent and cause of failure is determined. For operational reasons, individual tests may be continued pending investigation of a sampling test failure, but the final acceptance of product is contingent upon the inspector's decision regarding the overall conformance of the product to specification requirements. If investigation indicates that the defects may exist on items previously accepted, full particulars concerning the defects found, including recommendations for correction, will be furnished the procuring activity.

4.4.2.3 Valves selected for sampling tests shall be tested to determine compliance with the requirements specified in 3.4.1. The valve shall be placed 1 foot from the container and the container and valve shall be actuated at the same time. Any visual evidence of agent discharging from the nonactuated side of the valve shall constitute a failure. The actuating voltage shall be 18 volts momentarily applied.

* 4.4.2.4 Proof pressure test. Valves selected for sampling tests shall be hydrostatically tested at 400 pounds per square inch gage to determine proper strength of parts and materials when subjected to proof pressure. The pressure shall be applied through the inlet with the outlets capped. Pressure shall be applied at least two times for a period of 1 minute. The valve shall be actuated after each application of pressure and shall function satisfactorily. There shall be no evidence of failure, distortion, or permanent set.

* 4.5 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs shall be inspected to determine the preservation, packaging, packing and marking for shipment and storage conform to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A. Each valve assembly shall be packaged Method ICl, in accordance with MIL-P-116 (without preservative compound) and over boxed in a snug-fitting unit container conforming to PPP-B-676.

* 5.1.2 Level C. Packaging that will afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice, provided the latter meets the requirements of this level.

* 5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Valve assemblies packaged as specified in 5.1.1 shall be packed in exterior overseas type shipping containers conforming to PPP-B-636 or PPP-B-576. As far as practicable, exterior shipping containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each

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shipping container shall not exceed the limitations of the container specification. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers conforming to PPP-B-576 shall be provided with a case liner conforming to MIL-L-10547 and sealed in accordance with appendix thereto.

* 5.2.2 Level B. Valve assemblies packaged as specified in 5.1.1 shall be packed in domestic type exterior shipping containers conforming to PPP-B-636 or PPP-B-576. Exterior shipping containers shall be of minimum cube and tare consistent with the protection required. As far as practicable, exterior shipping containers shall be of uniform shape and size, and contain identical quantities. The gross weight of each shipping container shall not exceed the limitation of the container specification. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, such containers shall meet the special requirements table of PPP-B-636, as applicable.

* 5.2.3 Level C. Packing that will afford adequate protection at the lowest rate against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

* 5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type MC-1 valve covered by this specification is for use in a liquid agent aircraft fire extenguishing system to direct the extinguishing agent to any specified area.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of the specification.

b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.2.1 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Materiel Area, Service Engineering Division, ATTN: SANE, Kelly AFB, Texas 78241, and information pertaining to qualification of products may be obtained from that activity.

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6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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